Dipartimento: INGEGNERIA INFORMATICA, AUTOMATICA E GESTIONALE "ANTONIO RUBERTI"

Facoltà di Ingegneria dell'informazione, informatica e statistica, Università di Roma Sapienza
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Content

- Introduction
- Case study
- Proposed model
- Experiment solution
- Results
- Conclusions and future studies

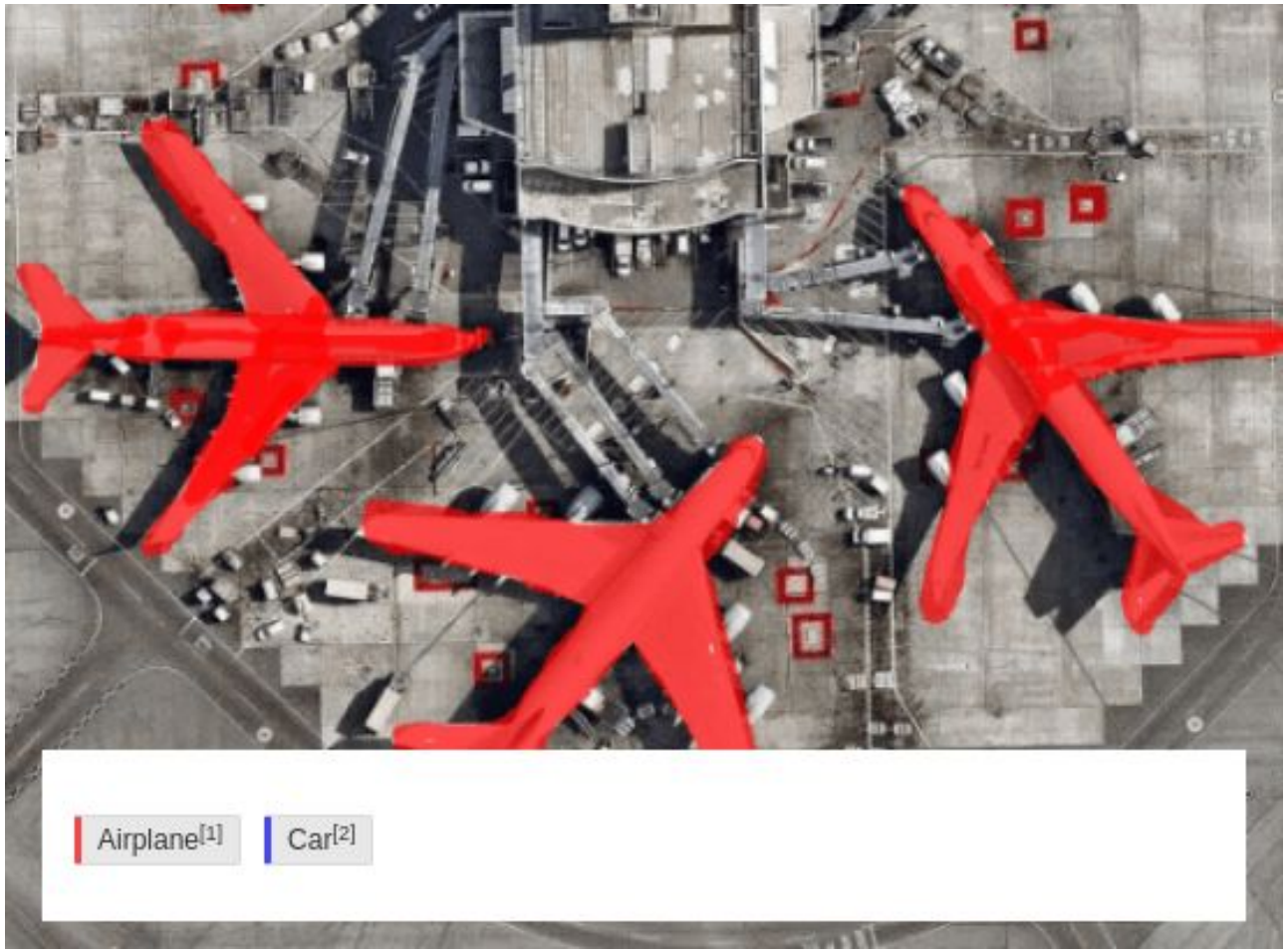


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Semantic Segmentation for Remote Sensing Images



Galileo satellite. E.S.A. project



Aerial Image with related segmentation mask in overlay.

It is shown how each pixel of the mask is labelled with the category of the object that represents.

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Remote Sensing Image Sample

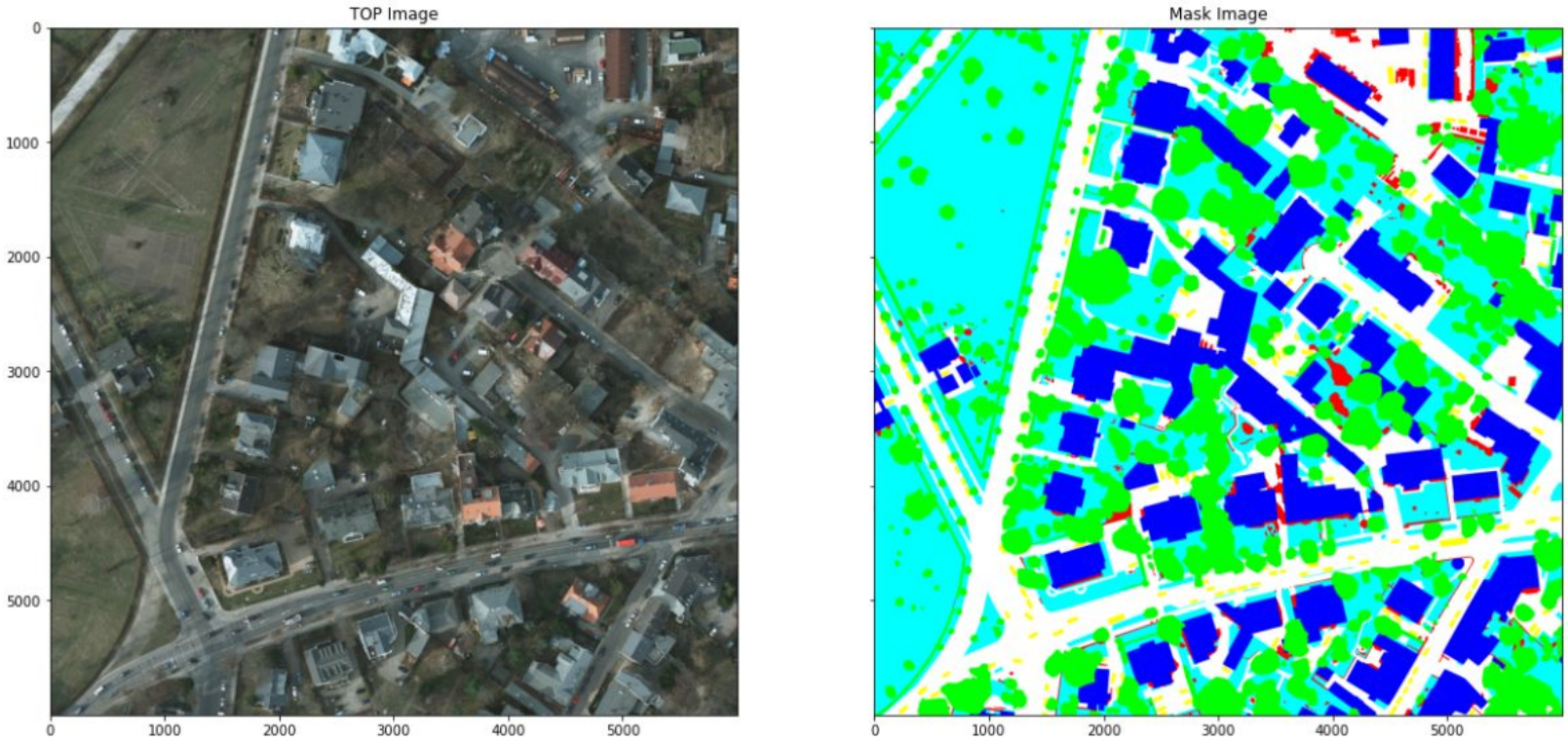


Image sample of a RSI with its mask from ISPRS Potsdam Dataset. The treue orthophoto on the right while the masks on the left. Each object in the scene is labelled with the category of interest.



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Dataset	images	task	References
Cityscapes	20K	Semantic segmentation	Daimler AG R&D, TU Darmstadt, MPI Informatics, TU Dresden
ADE20K	20K	Semantic segmentation, image to image transllation	University Alicante/Spain
COCO-stuff	164K	Object detection, instance segmentation, semantic segmentation	Microsoft
PASCAL Context	20K	Semantic segmentation	Standford University

Table 1.1. Most Popular and complete semantic segmentation datasets



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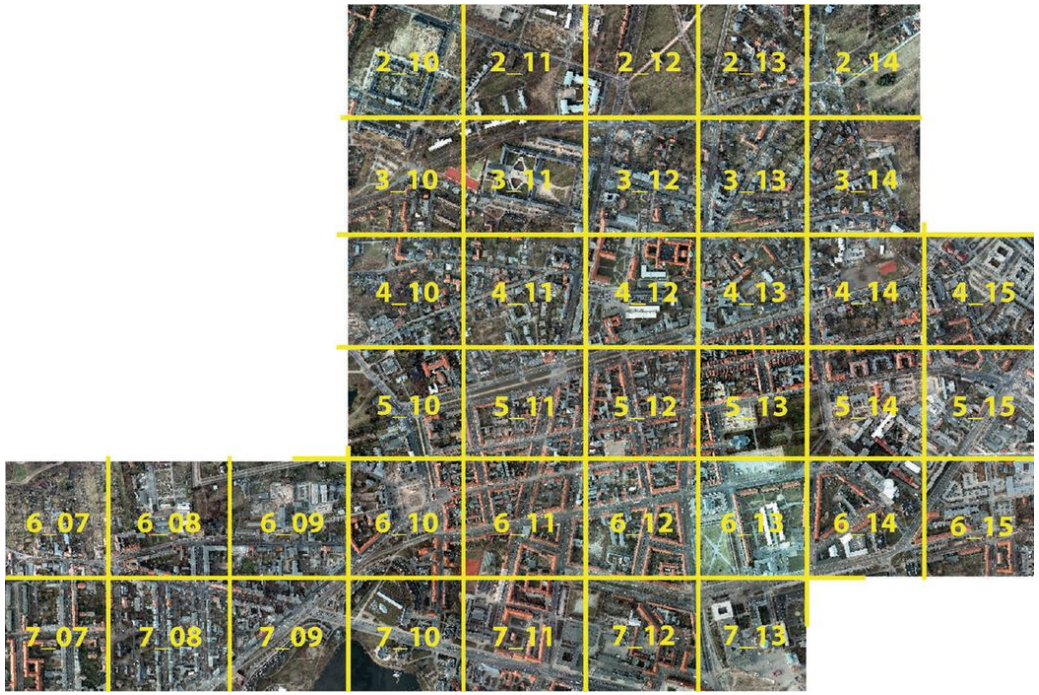






2D Semantic Labeling Dataset - city of Potsdam

The data set contains 38 patches (of the same size), each consisting of a true orthophoto (TOP) extracted from a larger TOP mosaic.



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Remote Sensing Image Sample

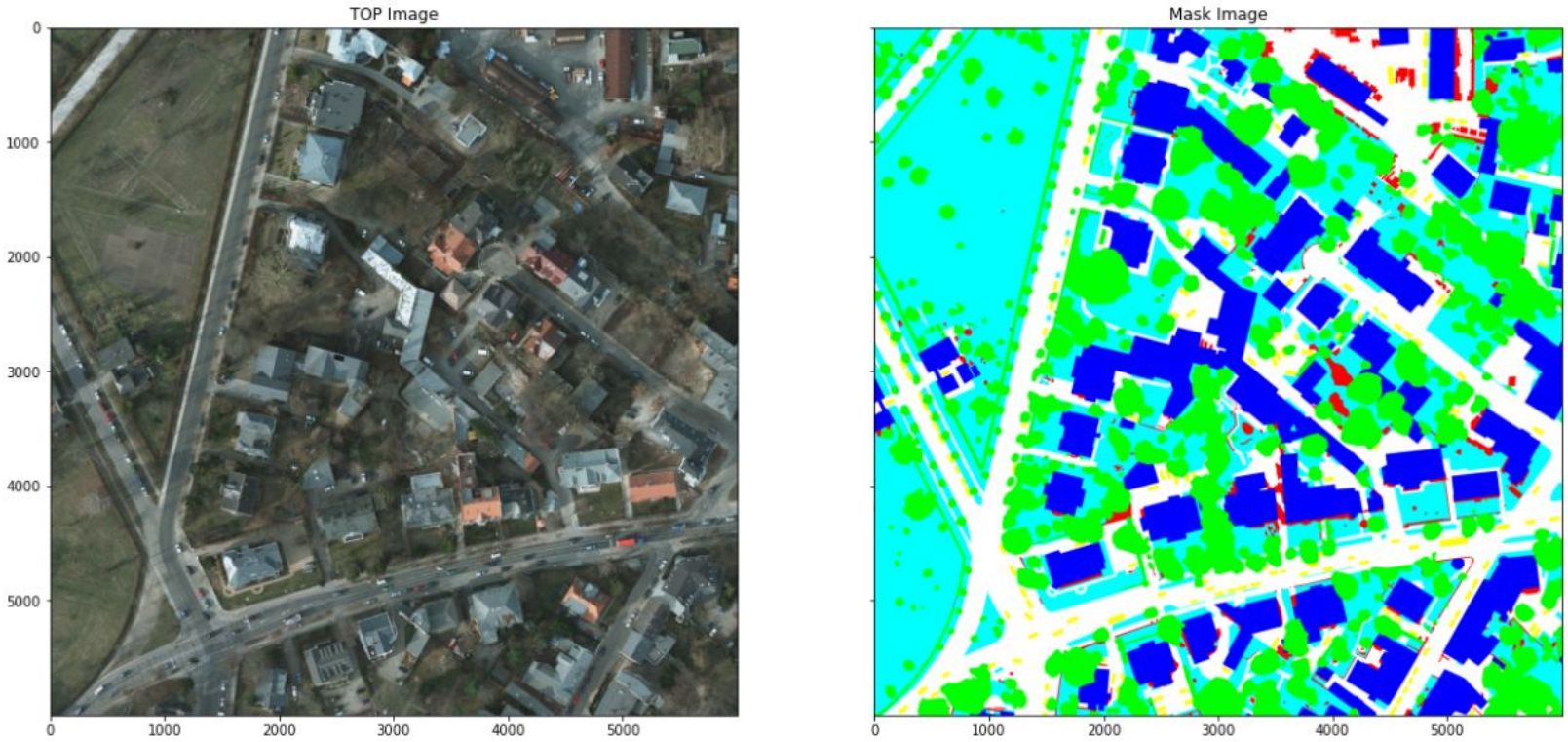
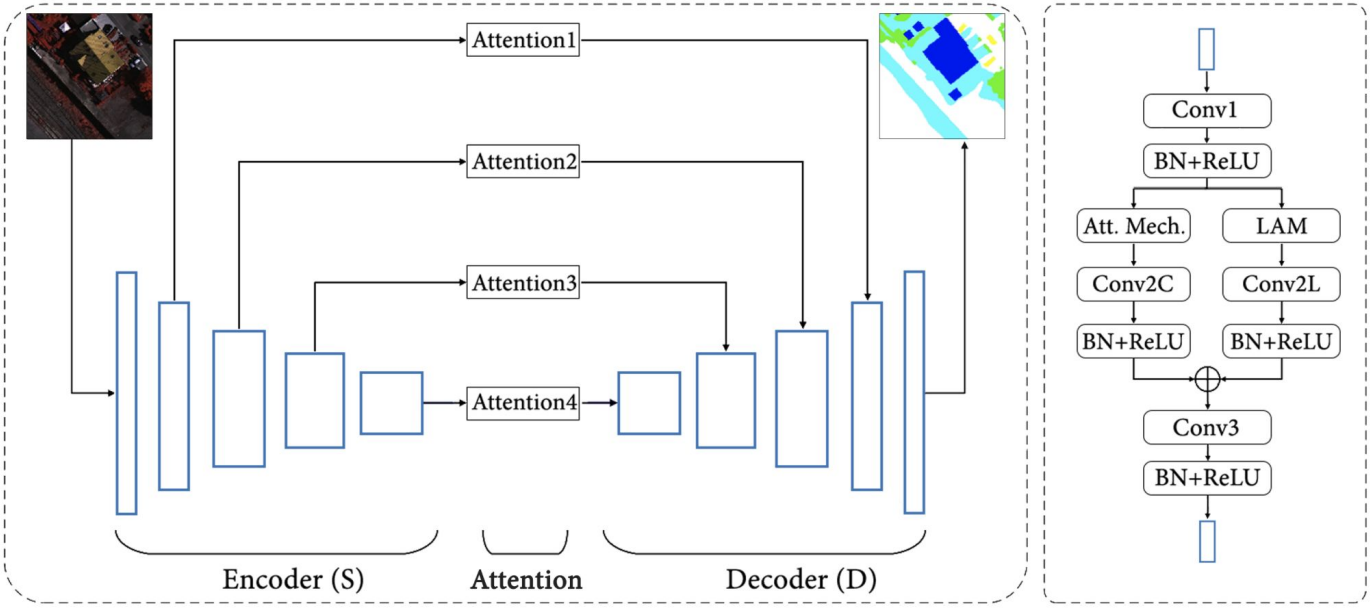


Image sample of a RSI with its mask from ISPRS Potsdam Dataset. The treue orthophoto on the right while the masks on the left. Each object in the scene is labelled with the category of interest.

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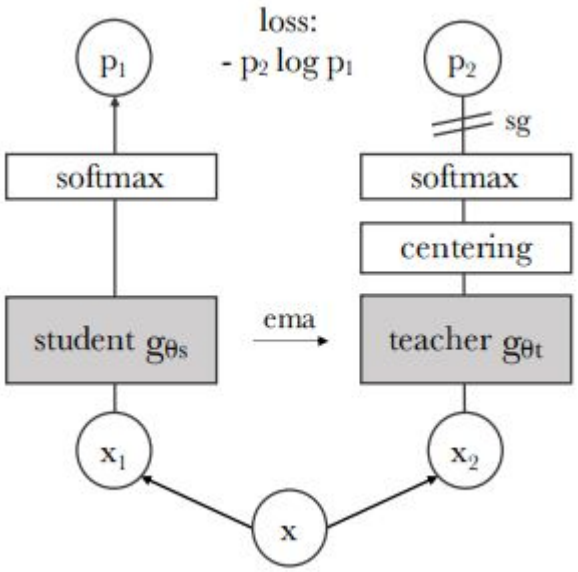
MAResUNet Architecture

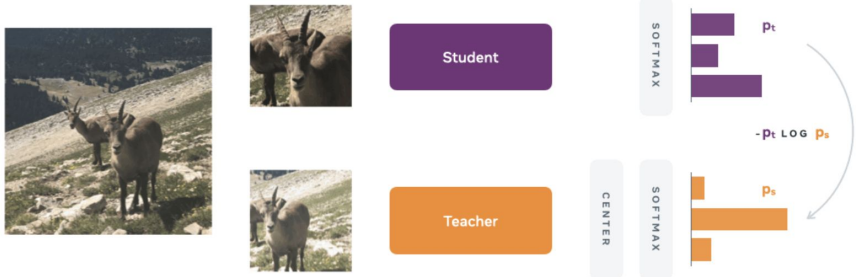


MARes-UNet architecture (left) and MARes-UNet attention block (right).

(MARes-UNet : Multi Stage Attention Residual Network)

DINO Pre-Training

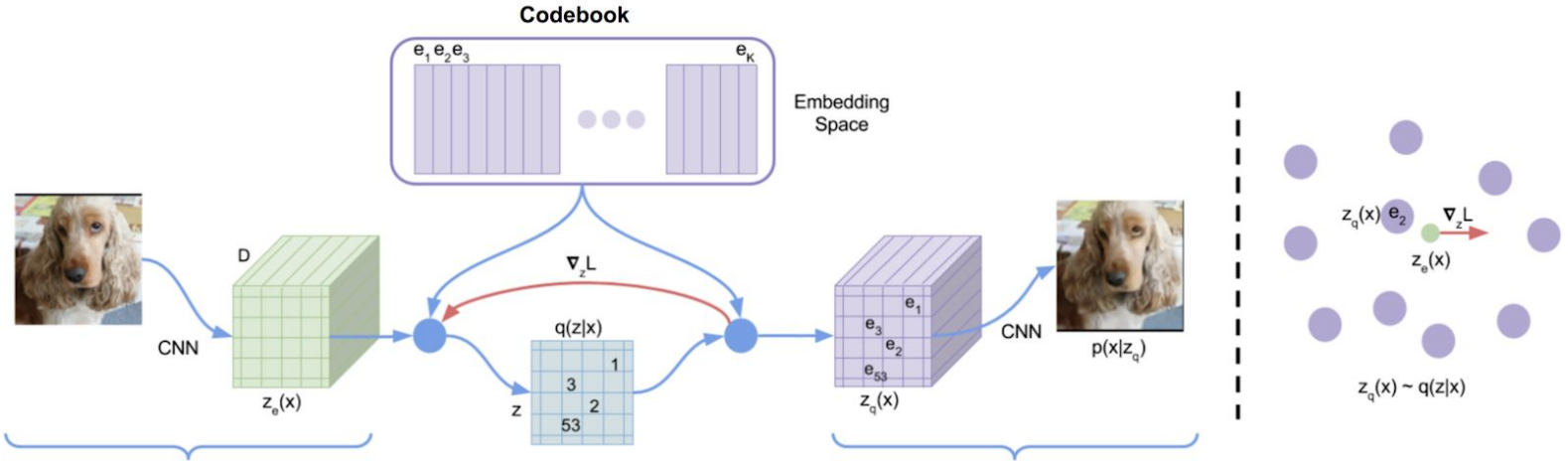




Both networks have the same architecture but different parameters. The output of the teacher network is centered on a mean computed over the batch. Each network outputs a K dimensional feature normalized with a temperature softmax over the feature dimension. Their similarity is then measured with a cross-entropy loss. We apply a stop-gradient (sg) operator on the teacher to propagate gradients only through the student. The teacher parameters are updated with an exponential moving average (ema) of the student parameters.

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VQ-VAE Architecture



$$q(\mathbf{z} = \mathbf{e}_k|\mathbf{x}) = \begin{cases} 1 & \text{if } k = \arg\min_i \|\mathbf{z}_e(\mathbf{x}) - \mathbf{e}_i\|_2 \\ 0 & \text{otherwise.} \end{cases}$$

Left: A figure describing the VQ-VAE. Right: Visualisation of the embedding space. The output of the encoder z(x) is mapped to the nearest point e2. The gradient Δz L (in red) will push the encoder to change its output, which could alter the configuration in the next forward pass. Source : "Neural Discrete Representation Learning"

$$L = \log p(x|z_q(x)) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta\|z_e(x) - \text{sg}[e]\|_2^2.$$



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Evaluated Models in Test Phase

Model	backbone	pretrain	OA	mIoU	F1
MAResUNet	ResNet18	ImageNet	79.05	60.44	72.16
MAResUNet	ResNet34	ImageNet	79.12	60.42	72.15
MAResUNet	ResNet18	DINO	74.14	52.95	65.65
MAResUNet	ResNet34	DINO	73.92	52.52	65.27
MAResUNet-VQ	ResNet18	ImageNet	74.99	54.27	66.83
MAResUNet-VQ	ResNet34	ImageNet	74.37	53.39	65.95
MAResUNet-VQ	ResNet18	DINO	70.22	46.47	59.28
MAResUNet-VQ	ResNet34	DINO	70.33	46.68	59.49

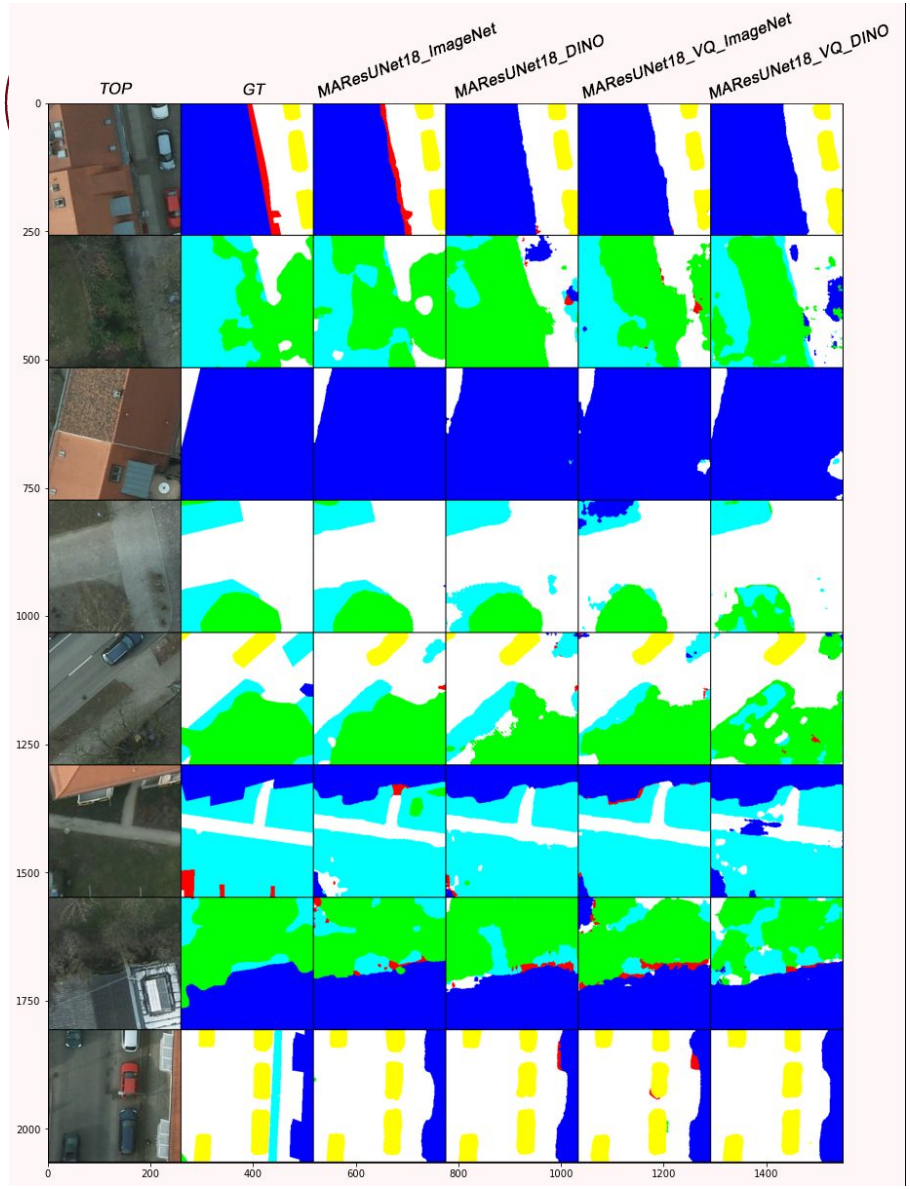


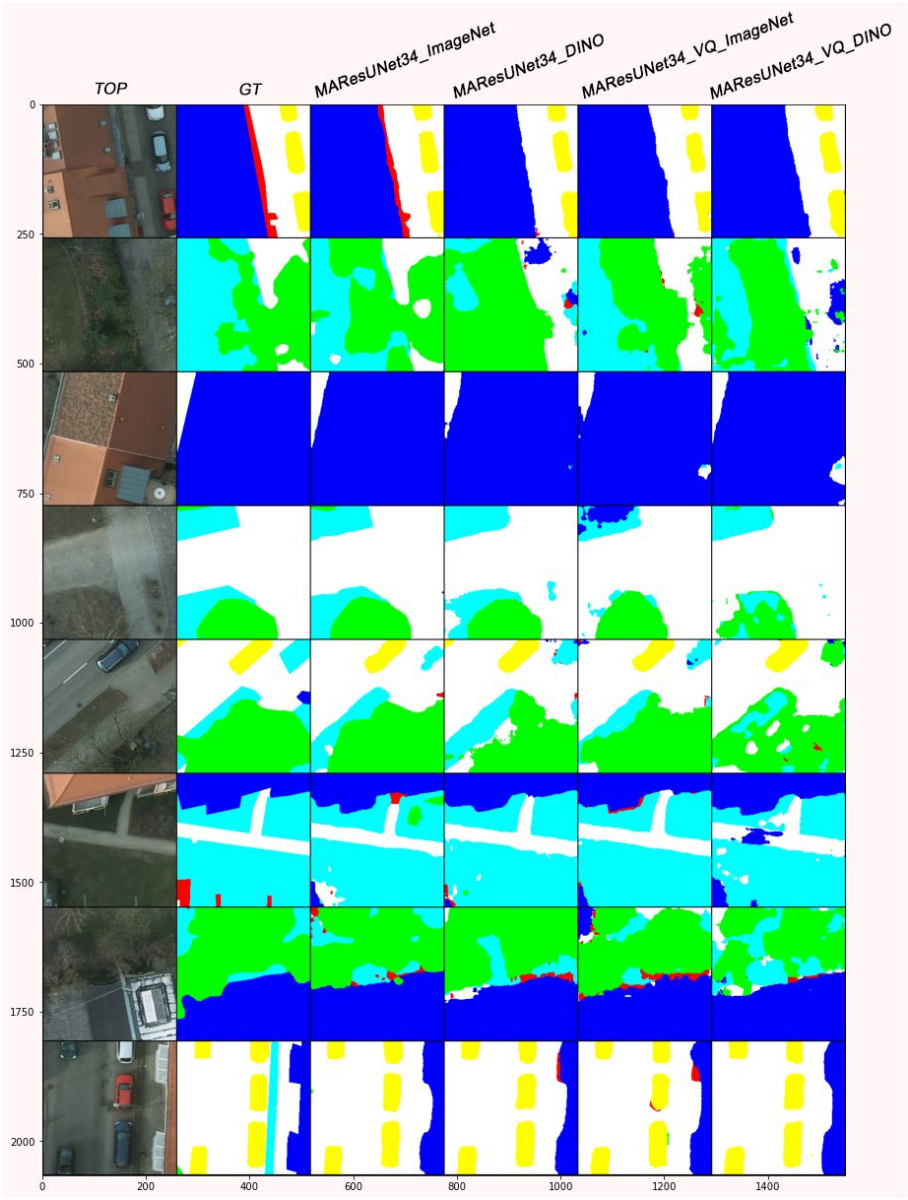
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Per Class Pixel accuracy scores from test phase

Model	backbone	pretrain	Accuracy per Class					
			Background	ImSurf	Car	Building	LowVeg	Tree
MAResUNet	ResNet18	ImageNet	**0.207613**	0.881230	**0.837966**	0.895138	**0.807360**	0.733300
MAResUNet	ResNet34	ImageNet	0.204663	**0.890064**	0.831177	0.894049	0.783754	**0.754808**
MAResUNet	ResNet18	DINO	0.114783	0.866064	0.766446	0.882792	0.673407	0.669019
MAResUNet	ResNet34	DINO	0.116300	0.858134	0.780328	0.887700	0.632859	0.705604
MAResUNet-VQ	ResNet18	ImageNet	0.108320	0.859215	0.797041	0.860000	0.725620	0.712245
MAResUNet-VQ	ResNet34	ImageNet	0.103615	0.867067	0.829848	0.872386	0.652675	0.723041
MAResUNet-VQ	ResNet18	DINO	0.035967	0.851734	0.732345	0.879178	0.570784	0.601483
MAResUNet-VQ	ResNet34	DINO	0.036300	0.843620	0.763357	**0.902357**	0.548373	0.609042

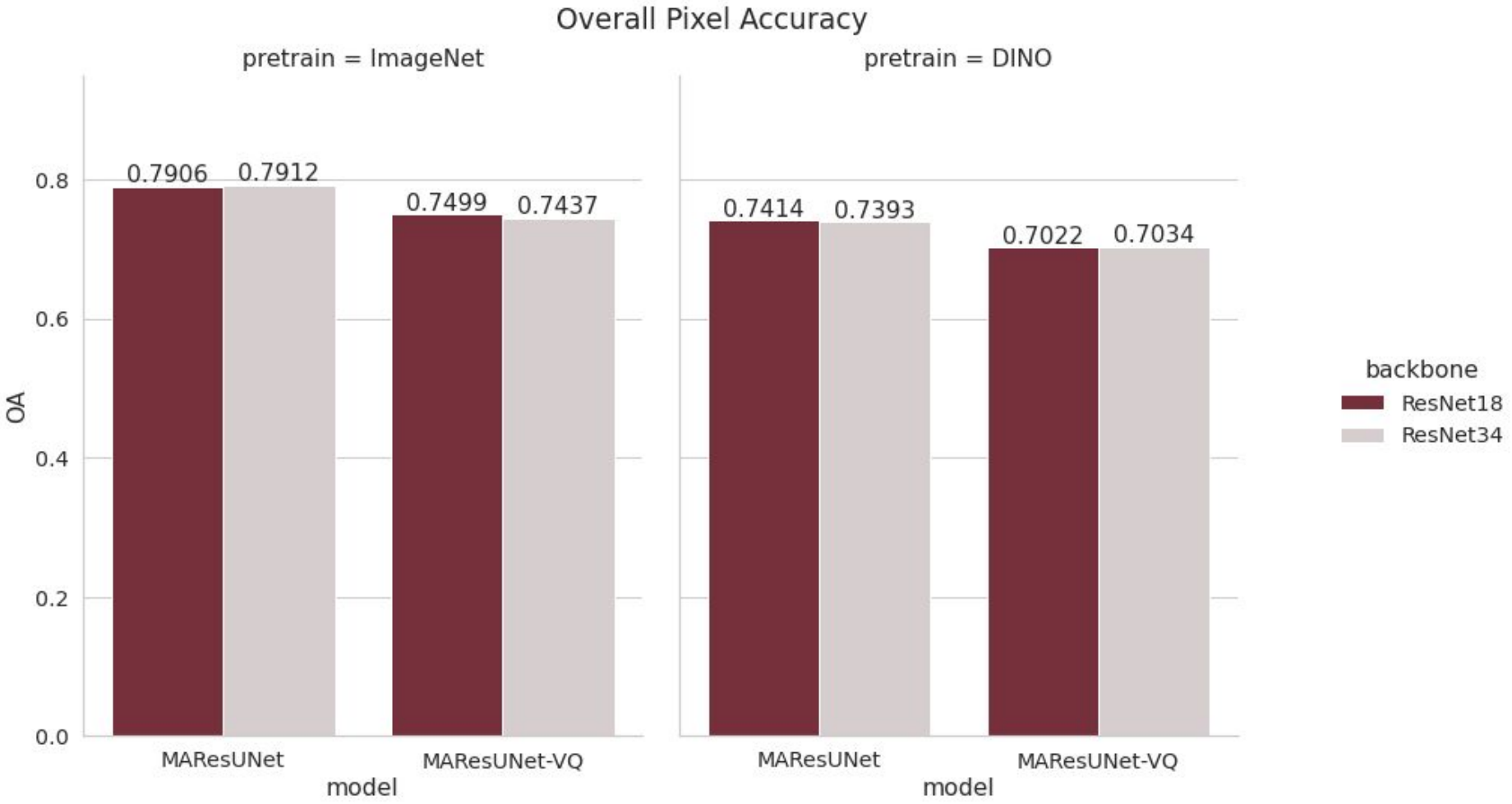
Table 4.3. Pixel accuracy scores from test phase.


TOP
GT
MAResUNet18_ImageNet
MAResUNet18_DINO
MAResUNet18_VQ_ImageNet
MAResUNet18_VQ_DINO
0
250
500
750
1000
1250
1500
1750
2000
0
200
400
600
800
1000
1200
1400


TOP
GT
MAResUNet34_ImageNet
MAResUNet34_DINO
MAResUNet34_VQ_ImageNet
MAResUNet34_VQ_DINO
0
250
500
750
1000
1250
1500
1750
2000
0
200
400
600
800
1000
1200
1400

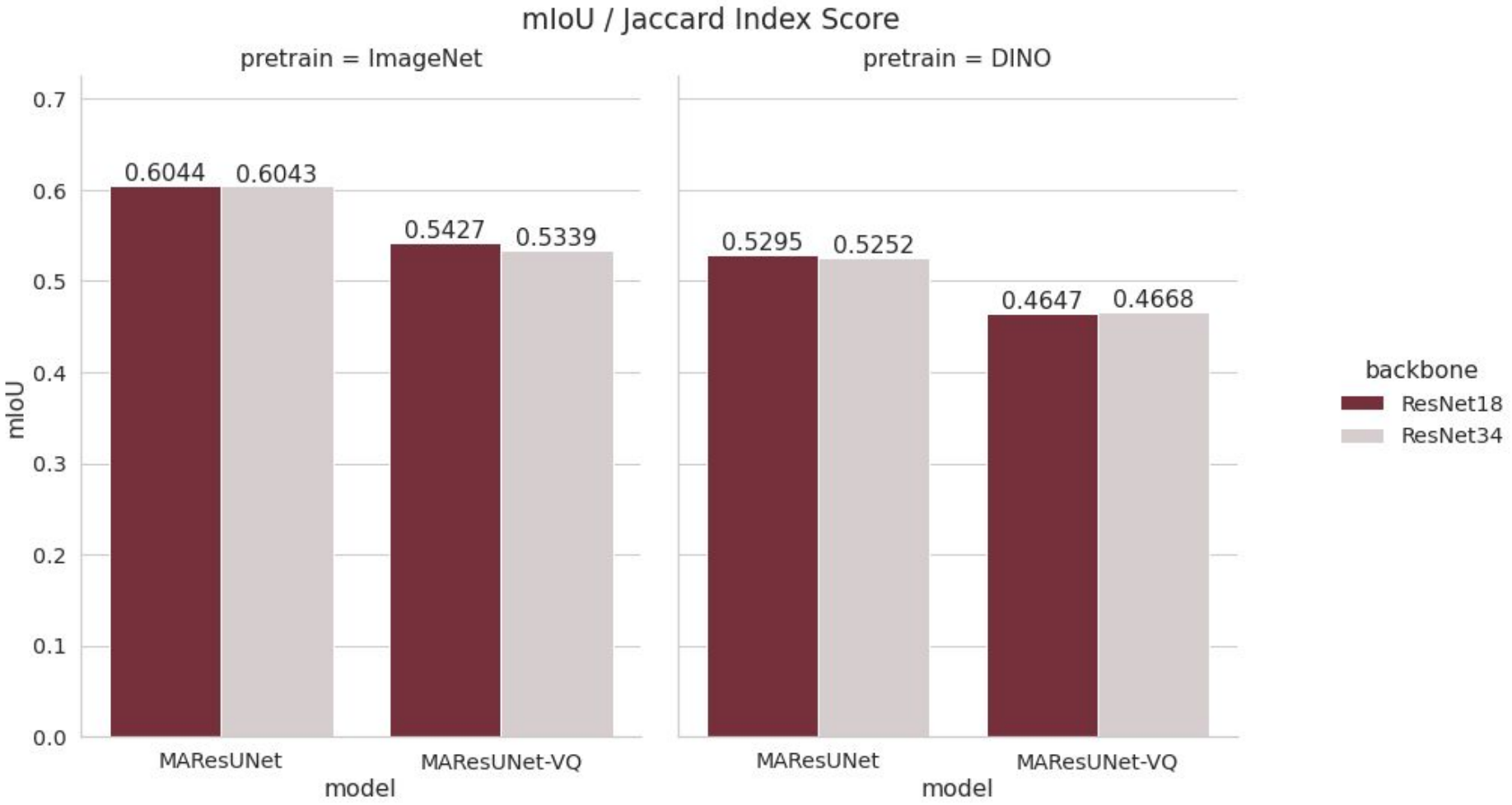
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Pixel accuracy scores per model



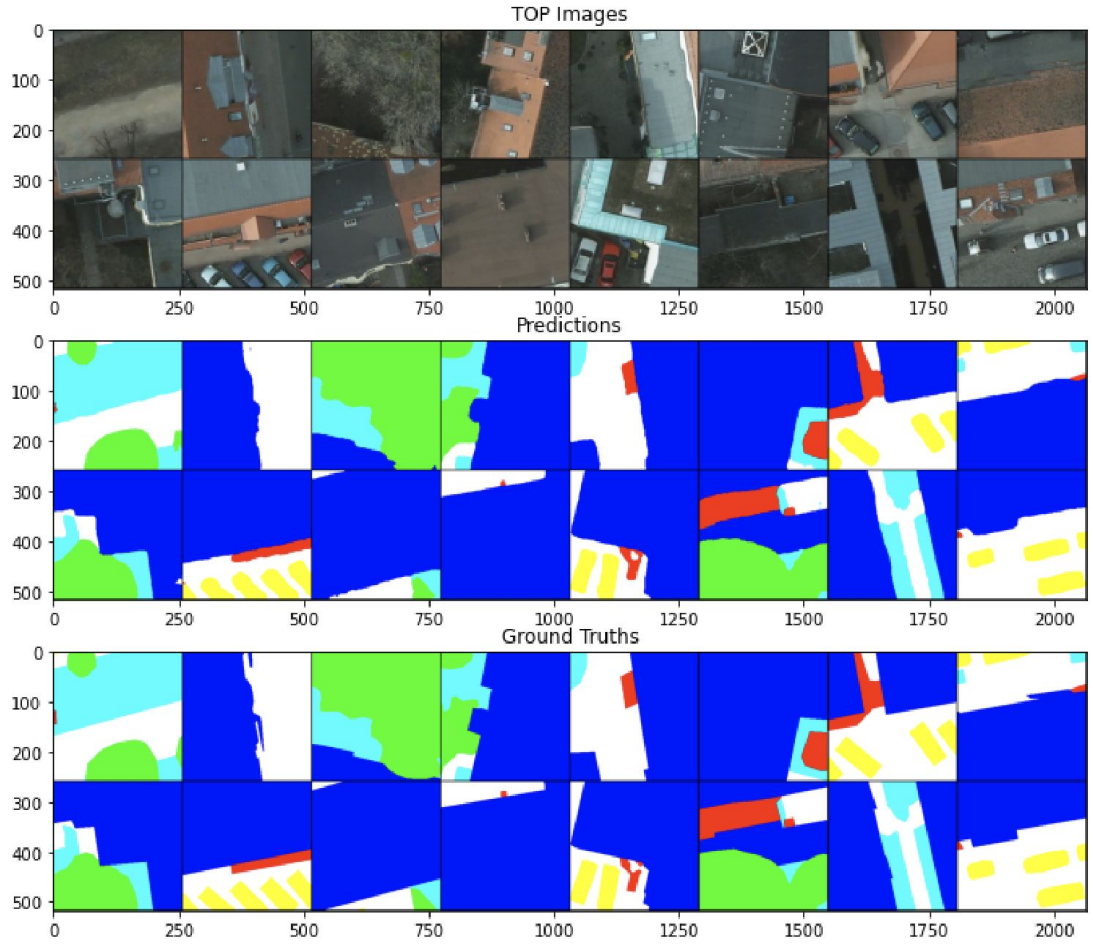
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mIoU/Jaccard Index scores per model



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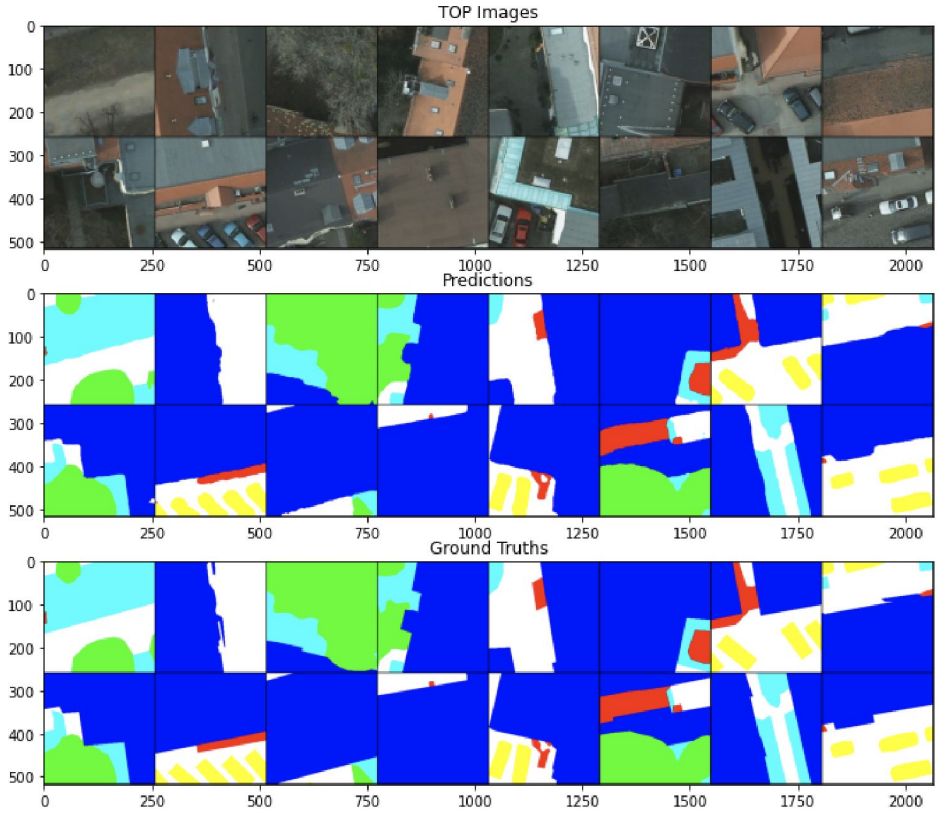
MAResUNet inference sample

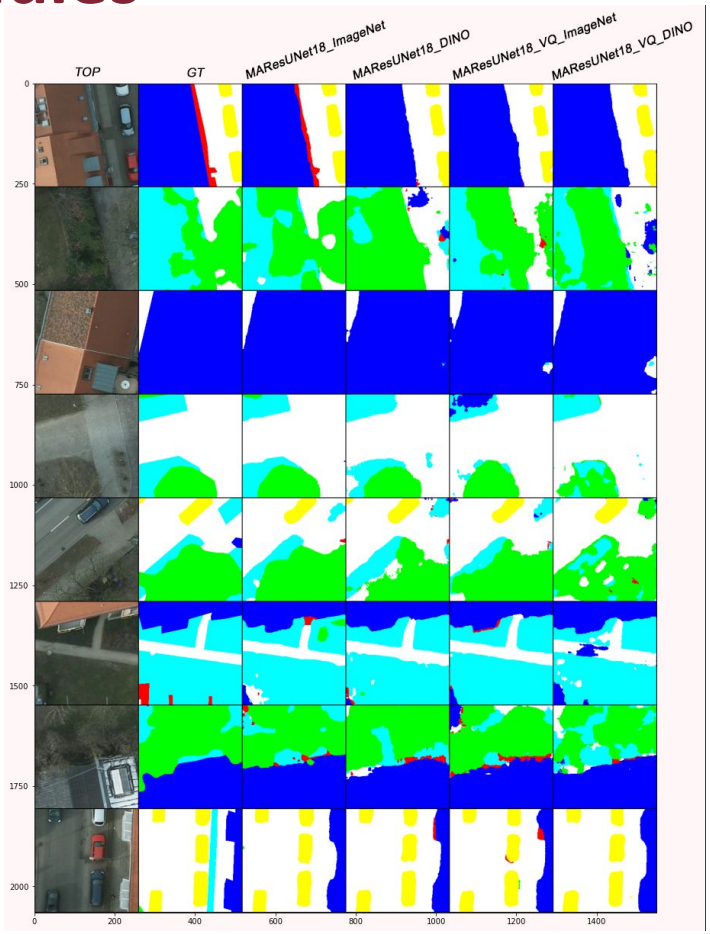


Batch of patches extracted with the use of the MAResUNet model using a ResNet 18 as backbone.

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Conclusion and future studies





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Grazie per la Vostra Attenzione!